

# M.N. NAFEEL

**Mechanical Engineering Technologist | Mechanical Assembly & CAD**

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Kandy, Sri Lanka (Willing to relocate to UAE immediately)

## SUMMARY

Mechanical Engineering Technology graduate with hands-on industrial workshop experience in machining, fitting, assembly, fabrication, component repair, and dimensional inspection. Proficient in creating mechanical parts and assemblies using SolidWorks, Fusion 360, and Autodesk Inventor; portfolio samples available online. Skilled in reading engineering drawings, applying geometric tolerances, and verifying dimensions with precision measuring instruments. Detail-oriented, with expertise in CAD, prototype development, small-batch manufacturing, 3D printing, and product finishing. Seeking to leverage technical and manufacturing skills in a dynamic engineering environment.

## EDUCATION

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| Bachelor of (Honors) in Mechanical Engineering Technology<br><i>Lincoln University College   Second Class Upper Division (GPA 3.38)</i>                            | 2024 - 2026 |
| City & Guilds Level 5 IVQ Advanced Technician Diploma in Mechanical Engineering<br><i>City &amp; Guilds of London Institute   <a href="#">View Credentials</a></i> | 2023 - 2025 |
| Diploma in Production Technology<br><i>Sri Lanka German Training Institute   Merit Pass</i>                                                                        | 2020 - 2023 |

## EXPERIENCE

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| <b>HEMAS Consumer Brands – Manufacturing Intern</b><br><b>Dankotuwa, Sri Lanka</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Nov 2025 – Apr 2026</b> |
| <ul style="list-style-type: none"><li>Identified equipment availability losses, speed losses, and quality defects across production lines through OEE and downtime analysis.</li><li>Supported process improvement initiatives in [product category, e.g., FMCG soap manufacturing] by reviewing production workflows, plant layouts, and GMP compliance.</li><li>Applied Lean Manufacturing, TPM, Kaizen, 5S, and Six Sigma fundamentals to support continuous improvement and operational efficiency.</li><li>Improved daily production monitoring and variance analysis by extracting production and material data from SAP.</li><li>Collaborated with production and maintenance teams to troubleshoot operational issues and prioritize improvement actions.</li><li>Supported the preparation and review of production data, performance reports, and improvement documentation.</li><li>Contributed to manufacturing operations focused on process consistency, workplace safety, quality standards, and production efficiency.</li></ul>     |                            |
| <b>Sri Lanka Ports Authority – Industrial Trainee</b><br><b>Colombo, Sri Lanka</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Sep 2022 – Mar 2023</b> |
| <ul style="list-style-type: none"><li>Machined mechanical components in a machine shop using CNC, conventional lathes, milling, drilling, and fitting operations.</li><li>Performed dimensional inspections of machined components using vernier calipers, micrometers, and other metrology tools.</li><li>Interpreted engineering drawings and GD&amp;T specifications to machine components within <math>\pm 0.01</math> mm tolerance.</li><li>Assisted with preventive and corrective maintenance, mechanical troubleshooting, welding, pipe installation, fabrication, and component repair.</li><li>Supported workshop personnel during maintenance and manufacturing activities involving industrial mechanical equipment.</li><li>Developed practical experience in machining, fabrication, engineering inspection, maintenance support, safe workshop practices, and technical problem-solving.</li><li>Maintained focus on dimensional accuracy, equipment condition, work quality, and compliance with engineering requirements.</li></ul> |                            |
| <b>Freelance CAD Engineer</b><br><b>Kandy, Sri Lanka</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Jan 2026 – Present</b>  |

- Develop parametric 3D models, assemblies, and engineering drawings in Autodesk Inventor, SolidWorks, Fusion 360, and AutoCAD.
- Reverse engineer existing components by analyzing geometry, dimensions, interfaces, clearances, and functional requirements.
- Apply machine design, GD&T, DFM, and DFA principles to improve manufacturability, assembly, and serviceability.
- Use assembly design and motion analysis to evaluate component relationships, clearances, degrees of freedom, and mechanical motion.
- Develop Python and Build123D workflows to automate model generation and support design iterations.
- Prepare technical design documentation for manufacturing, prototyping, and design review.
- Collaborate with clients to clarify requirements, review design alternatives, and incorporate engineering feedback.

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## SKILLS

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Parametric 3D CAD Modeling | Finite Element Analysis (FEA) | Mechanical Design | Machine Design | Mechanical Systems Design | Assembly Design | Reverse Engineering | Engineering Drawings | Geometric Dimensioning and Tolerancing (GD&T) | Design for Manufacturability (DFM) | Design for Assembly (DFA) | Sheet Metal Design | Surface Modeling | Motion Analysis | Design Validation | Prototype Development | Manufacturing-Focused Design | Component Design | Technical Documentation | Design Optimization | CAD Automation

**Software:** SolidWorks | Fusion 360 | Autodesk Inventor | AutoCAD | MATLAB | Ansys

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## PROJECTS

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### Feasibility Analysis of a Small-Scale Plastic Pyrolysis System

2025 – 2026

#### Final-Year Academic Project

- Designed a 30-liter batch plastic pyrolysis system in SolidWorks, including an AISI 304 reactor vessel, an external heating system, a vapor piping network, and a shell-and-tube condenser assembly.
  - Performed heat transfer calculations, material selection analysis (AISI 304 vs. alternatives), thermal load estimation, and condenser sizing based on vapor flow and cooling requirements. Evaluated thermal performance, operating constraints (temperature, pressure), safety considerations (pressure relief, thermal expansion), and system scalability for larger batch sizes.
  - Completed a financial feasibility analysis, including capital cost estimation, operating expenses, projected revenue from pyrolysis oil, and payback period calculation.
  - Developed a mechanical system concept integrating equipment layout, process flow, piping connections, heating requirements, and operational considerations.
  - Documented design assumptions, engineering constraints, calculations, and feasibility findings using technical reporting methods.
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## CERTIFICATIONS & PROFESSIONAL AFFILIATIONS

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**Certifications:** Occupational Safety and Health | Chemical Handling Health & Safety | Firefighting & Rescue Training | Machine Learning for Engineers

**Affiliations:** Affiliate Student Member, Institution of Mechanical Engineers (IMechE) | Member, International Association of Engineers (IAENG)

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## LANGUAGES

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English (Working Proficiency) | Arabic (Basic) | Tamil (Native) | Sinhala (Conversational)

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## ADDITIONAL INFORMATION

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Interested in hands-on workshop production, prototyping, product assembly, finishing, and quality control. Willing to learn FDM/SLA printing workflows, slicing software, mould making, and casting processes.

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## REFERENCES

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- Available upon request.